

## §12.6 Cylinders and Quadrate Surfaces

**Homework :** 9,21-28,29,31,35,43,46,48

- (1) Cylinders : A surface consisting of all lines (called rulings) that are // to a given line and pass through a given plane curve.

**Example 1 :** (i)  $z = x^2$  (Parabolic cylinder)

(ii)  $x^2 + y^2 = 1$

**Remark :** If one of the variables  $x$ ,  $y$  or  $z$  is missing from the equation of a surface, then the surface is a cylinder.

- (2) Quadratic Surfaces :

$$ax^2 + by^2 + cz^2 + dxy + eyz + fxz + gx + hy + iz + j = 0$$

See P.836 of the test book for the classification of quadratic surface.

- (3) Cross-sections or traces